EFY TESTED S.C. Dwivedi

AUTOMATIC WASH BASIN TAP
With LED Light

SURESH CHANDRA DWIVEDI

When waking up at night to use the wash basin, you may struggle to find the switchboard to turn on the light before opening the wash basin tap to wash your hands. To help, this automatic wash basin tap with built-in LED light has been designed using an IR module and 555 timer IC. The IR module is a very simple and compact device, which has transmitting and receiving IR LEDs and a comparator.

This device opens the wash basin tap automatically and switches on the LED light fitted above the wash basin when one reaches in front of the wash basin. The wash basin tap and the LED light turn off automatically after a set time.

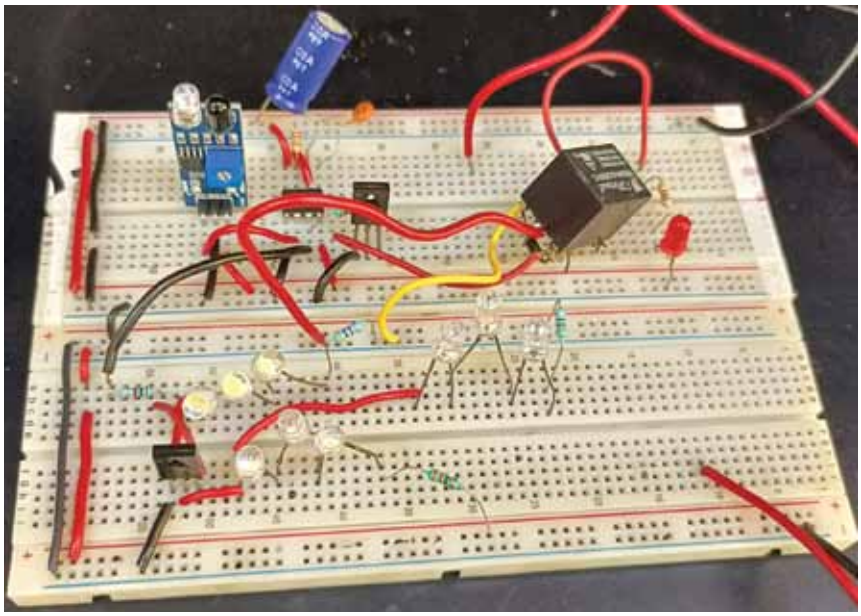

Fig. 1: Author's prototype

Circuit and working

The circuit diagram of the device is shown in Fig. 2. It is built around a 230V AC primary to 12V, 500mA secondary transformer (X1), a bridge rectifier (BR1), an IR module connected across CON2, 5V voltage regulator LM7805 (IC1), timer NE555 (IC2), and a few other components.

The 230V AC mains is stepped down to 12V through the step-down transformer X1. This 12V secondary output of X1 goes to the bridge rectifier BR1. Capacitor C1 filters the ripples in voltage and the rectified output is given to regulator 7805 (IC1). IC1 provides 5V regulated DC

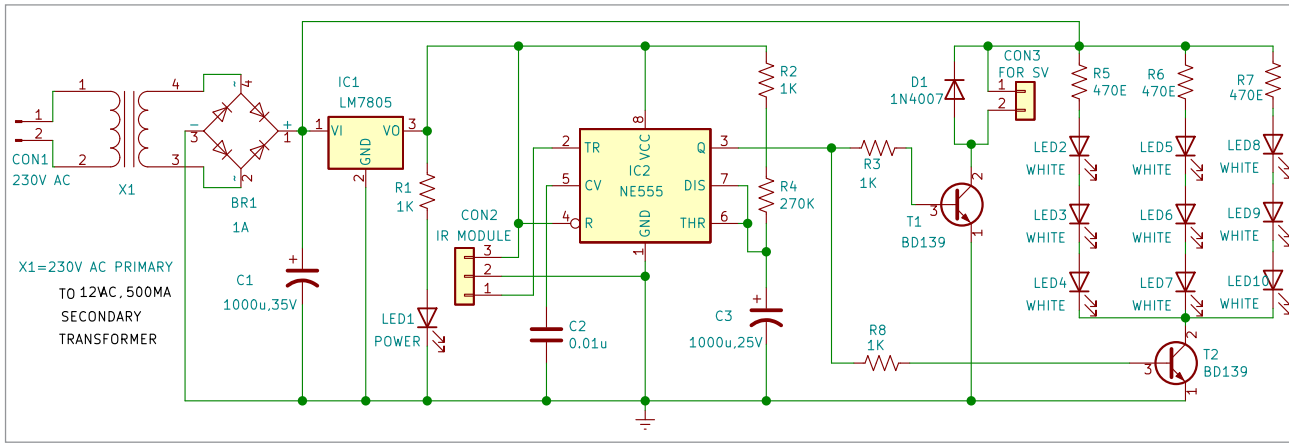

Fig. 2: Circuit diagram